Laplacian Eigenvectors of Non-uniformly Dense Random Circulant Graphs

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Abstract

Let G be a circulant graph with a vertex set $V = \{0, 1, \ldots, n-1\}$, and connect every vertex i to vertices $(i+k)$ and $(i-k) \mod n$ for each k in a set $S \subset \{1, 2, \ldots, \lfloor \frac{n}{2} \rfloor\}$. The eigenvectors of a graph's Laplacian matrix can be used to generate an optimal layout that minimizes the distances between connected vertices [4]. For circulant graphs, this layout naturally places vertices along a circle. In this thesis, we introduce the Laplacian eigenvectors of circulant graphs and study how well this property persists when we introduce randomness to the graph structure. Specifically, we investigate non-uniformly dense random graphs with underlying circulant structure. We introduce a model graph that approximates the non-uniformly dense random graph and propose a possible structure for its eigenvectors.

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1 Introduction

1.1 Motivation

The Laplacian eigenvectors of graphs are a powerful tool for visualising and understanding their structure. Finding a suitable way of visualising an unknown graph can offer intuition and insight into analysing its structure and properties.

Naturally, we may want to find a way to visualise graphs such that vertices that are connected in a graph are close to each other. For a 1-dimensional visual, we may consider assigning each vertex a value on the real number line; we call this assignment an *embedding*. In order for connected vertices, or *neighbours*, to be close to each other, the edges between them should be as short as possible. We face an optimization problem: How can we assign values to the vertices such that we minimize the distances between them?

One can naturally extend this inquiry: What would be the optimal way to visualise a graph in two dimensions? What might the resulting embedding reveal about the graph's properties?

In this research paper, we investigate this question in the context of random circulant graphs. Our goal is to recover information about the underlying circulant model of randomly generated graphs by studying their embeddings.

1.1.1 Definitions

Definition 1. Graph

A graph G is defined by a set of vertices V and a set of edges E. We will focus on simple graphs, in which every edge connects two vertices $u, v \in V$ such that $u \neq v$. The edges are undirected: an edge (u, v) is equivalent to the edge (v, u). We denote two connected vertices u and v by $u \sim v$.

Definition 2. Adjacency Matrix

An adjacency matrix is a way of encoding the structure of a graph into an $n \times n$ matrix. We label the vertices by $v_0, v_1, \dots v_{n-1}$. The adjacency matrix A of a graph is then defined by

$$A_{ij} = \begin{cases} 1 & \text{if } (v_i, v_j) \in E \\ 0 & \text{else} \end{cases}$$

Since $(v_i, v_j) = (v_j, v_i)$, the matrix is by construction symmetric. There are no edges from a vertex to itself, so the diagonal is all zeros.

Definition 3. Laplacian Matrix

A Laplacian matrix is another way to represent a graph $G = (V, E)$ as a matrix. We define its

Laplacian matrix L by

$$L_{ij} = \begin{cases} \deg(v_i) & \text{if } i = j \\ -1 & \text{if } (v_i, v_j) \in E \\ 0 & \text{else} \end{cases}$$

where $\deg(v_i)$ is the number of vertices the vertex u is connected to by an edge.

Let x be a vector assigning values to the vertices of the graph. For any fixed vertex $u \in V$,

$$(Lx)(u) = \sum_{u \sim v} (x(u) - x(v)).$$

It follows that the quadratic form of L is

$$\begin{aligned} x^T L x &= \sum_{u \in V} x(u) \sum_{u \sim v} (x(u) - x(v)) \\ &= \sum_{u \in V} \left(\deg(u)\, x(u)^2 - \sum_{u \sim v} x(u)x(v) \right) \\ &= \sum_{u \in V} \deg(u)\, x(u)^2 - 2 \sum_{(u,v) \in E} x(u)x(v) \\ &= \sum_{(u,v) \in E} x(u)^2 - 2x(u)x(v) + x(v)^2 \\ &= \sum_{(u,v) \in E} (x(u) - x(v))^2 \geq 0. \end{aligned} \tag{1}$$

This implies that L is a positive semi-definite matrix, and thus all of its eigenvalues are non-negative. We order the eigenvalues of L from smallest to largest, where $\lambda_0 \leq \lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_{n-1}$. By construction, the row sum of every row of any Laplacian matrix L is 0. In particular, this implies that

$$L\mathbf{1} = \mathbf{0} = 0 \cdot \mathbf{1}. \tag{2}$$

So $\mathbf{1}$ is always an eigenvector of L with eigenvalue $\lambda_0 = 0$.

Definition 4. Circulant graph

A circulant graph can be described as a graph in which the vertices are placed along a circle and every vertex is connected to some fixed number d of its nearest vertices in both directions along the circle. Formally, if we number the vertices $0, 1, \ldots, n-1$ in their order around the circle, a vertex k will be adjacent to vertices $(k+s) \bmod n$ and $(k-s) \bmod n$ for every $s = 1, 2, \ldots, d$. The adjacency matrix of a circulant graph has the form

$$A = \begin{bmatrix} 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 & 0 & 0 & 1 \\ 1 & 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 & 1 & 0 & 1 \\ 1 & 1 & 0 & 0 & 1 & 1 & 0 \end{bmatrix}$$

where the main diagonal is bordered by diagonals with all 1s.

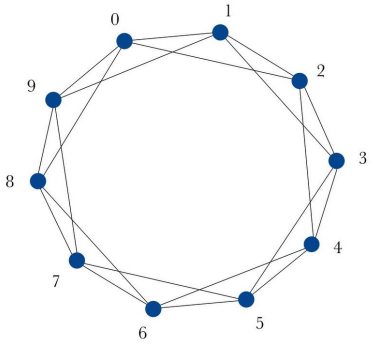


Figure 1: A circulant graph, $n = 10, d = 2$

2 Problem Statement

2.1 Visualising Graphs

Let us return to our initial question, and consider a graph G with n vertices. How can we draw the vertices of G such that the distances between the vertices are minimized?

Let $x \in \mathbb{R}^n$ be a vector assigning each vertex a value on the real number line. The length of an edge between vertices u and v will be the distance of their assigned values on the number line, that is, $|x(u) - x(v)|$. Hall [4] proposed that a suitable criterion of optimality for such an assignment is minimizing the sum of the squared distances between the vertices. Formally, this means finding x such that

$$x = \arg\min_{x} \sum_{(u,v) \in E} (x(u) - x(v))^2.$$

To avoid the trivial solution $x(u) = 0 \; \forall \; u \in V$, we restrict the solutions to $||x||^2 = 1$. This provides proper scaling for our embedding. By equation 1, the Laplacian matrix L of a graph is defined such that for a vertex assignment x,

$$x^T L x = \sum_{(u,v) \in E} (x(u) - x(v))^2.$$

Our optimization problem becomes finding x such that

$$x = \arg\min_{||x||=1} \sum_{(u,v) \in E} (x(u) - x(v))^2 = \arg\min_{||x||=1} x^T L x.$$

The Courant-Fischer Theorem, detailed in for example [9], tells us that for $i = 1, \ldots, n$:

$$\psi_i = \arg \min_{\substack{||x||=1 \\ x^T \psi_j = 0 \text{ for } j<i}} x^T A x.$$

The theorem implies that the first Laplacian eigenvector is a minimizer of $x^T Lx$ subject to $||x|| = 1$. By equation 2, the normalized first Laplacian eigenvector of any graph is

$$\phi_1 = \frac{1}{\sqrt{n}} \cdot \mathbf{1}.$$

With this assignment, all the vertices are plotted to the same coordinate and the length of every edge is 0. To avoid this, we only consider vectors orthogonal to $\mathbf{1}$. Again by the Courant-Fischer theorem, the second Laplacian eigenvector is a minimizer of

$$x = \arg\min_x x^T Lx = \arg\min_x \sum_{(u,v)\in E} (x(u) - x(v))^2.$$

where the optimization problem is subject to $||x|| = 1$ and $x^T\mathbf{1} = 0$. In particular, this implies that $x = \phi_2$ is a solution to our 1-dimensional visualisation problem. In practice, assigning the vertices in a graph values according to its second Laplacian eigenvector results in an optimal embedding of the graph on a real number line.

We can extend this to 2 dimensions. Let x and y be n-dimensional vectors assigning coordinates to the vertices of a graph G. The distance between two vertices in the 2-dimensional plane is

$$|(x(u), y(u)) - (x(v), y(v))| = \sqrt{(x(u) - x(v))^2 + (y(u) - y(v))^2}.$$

Minimizing the sum of the squared distances between all the vertices is equivalent to finding x and y such that

$$\begin{aligned} x, y &= \arg\min_{x,y} \sum_{(u,v)\in E} (x(u) - x(v))^2 + (y(u) - y(v))^2 \\ &= \arg\min_{x,y} (x^T Lx + y^T Ly). \end{aligned}$$

Like in the 1-dimensional case, we restrict our solutions to

$$||x|| = ||y|| = 1, \quad x^T\mathbf{1} = y^T\mathbf{1} = 0.$$

To avoid the trivial solution $x = y = \phi_2$, we add the additional constraint

$$x^T y = 0.$$

Then our problem becomes finding

$$x = \arg\min_{\substack{||x||=1 \\ x^T\mathbf{1}=0}} x^T Lx, \quad y = \arg\min_{\substack{||y||=1 \\ y^T\mathbf{1}=y^T x=0}} y^T Ly.$$

By the Courant-Fischer Theorem, we find the solution

$$x = \phi_2, \quad y = \phi_3.$$

So plotting the vertices of a graph in a 2-dimensional plane with coordinates according to the graph's second and third eigenvector

$$(\phi_2(i), \phi_3(i)) \quad \text{for } i = 0, 1, \ldots, n$$

results in a structure that minimizes the total length of all the edges between the vertices. This is true in general for every graph. A particularly interesting case is a circulant graph.

2.2 Background Proposition

Let L be the Laplacian matrix of a circulant graph of order n. We order the eigenvalues of L from smallest to largest, where

$$\lambda_1 = 0 \leq \lambda_2 \leq \lambda_3 \leq \cdots \leq \lambda_n,$$

with their respective eigenvectors $\psi_1, \psi_2, \ldots, \psi_n$. The first eigenvalue 0 and eigenvector $\mathbf{1}$ are common to every Laplacian matrix (eq. 2), and are therefore not very interesting when studying the individual properties of different types of graphs. We direct our focus to the non-zero eigenvalues and their eigenvectors.

We will prove that if we take the coordinates of the second and third eigenvectors of L, ψ_2 and ψ_3, the resulting points

$$(\psi_2(i), \psi_3(i)) \quad \text{for} \quad i = 0, 1, \ldots, n-1$$

lie on a common circle in the 2-dimensional plane.

Moreover, the ordering of the vertices in the original graph is preserved in this embedding: as i increases from 0 to $n-1$, the points $(\psi_2(i), \psi_3(i))$ traverse the circle in a cyclic order, making one full rotation around the circle. Moving from vertex i to $i+1$ corresponds to a small uniform rotation along the circle, maintaining the original vertex sequence.

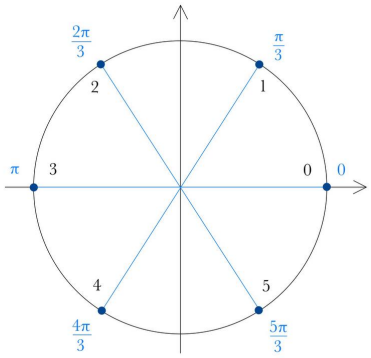


Figure 2: Example embedding a circulant graph, $n = 6$

We will see a proof of this proposition in Section 3.

2.3 Problem Statement

In their paper, Richter and Rocha [8] show that if we remove edges from a circulant graph independently with uniform probability, plotting the points $(\psi_2(i), \psi_3(i))$ for vertices $i = 0, 1, \ldots, n-1$ of the resulting random graph will still result in a mostly circular embedding. More importantly, they show that the order of the vertices in the embedding will be only slightly perturbed from that of the original circulant graph. This implies that with high probability, we can recover the original circulant graph by using the 2nd and 3rd Laplacian eigenvectors of the random graph.

Motivated by the result, we wanted to investigate how much can we perturb the structure of the

circulant graph until we lose the circular embedding and are no longer able to recover the original circulant graph by using the Laplacian eigenvectors of the random graph.

The circulant graph is very uniform; every vertex has the same degree. By removing edges randomly with uniform probability, the degrees of the vertices are no longer identical but probabilistically will be relatively uniform. Many real-world networks that are modelled by graphs will, however, have different areas with varying densities: some parts of the network will be very densely connected, while other parts will be sparser. We are interested in accounting for such fluctuating densities in our random graph model. Ultimately, we aim to find a way to retrieve an underlying order in such graphs.

We approach the problem by creating a model to generate non-uniformly dense random graphs with an underlying circulant structure. We generate a variety of such random graphs, compute their Laplacian eigenvectors, and visualise their embeddings. We then analyse the structure of the eigenvectors and aim find an explicit description of their structure. Finding such a description would allow us to prove that plotting the 2nd and 3rd Laplacian eigenvector of non-uniformly dense random graphs does, in fact, allow us to recover the original circulant structure.

3 Laplacian Spectrum of Circulant Graphs (Proof of Proposition)

3.1 Additional Definitions

Definition 5. The General Circulant Matrix

Every circulant matrix is uniquely defined by its first row $\mathbf{a} \in \mathbb{R}^n$, $\mathbf{a} = (a_0, a_1, \ldots, a_{n-1})$. The corresponding circulant matrix is

$$\operatorname{circ}(\mathbf{a}) = \begin{pmatrix} a_0 & a_1 & a_2 & \cdots & a_{n-2} & a_{n-1} \\ a_{n-1} & a_0 & a_1 & \cdots & a_{n-3} & a_{n-2} \\ a_{n-2} & a_{n-1} & a_0 & \cdots & a_{n-4} & a_{n-3} \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ a_1 & a_2 & a_3 & \cdots & a_{n-1} & a_0 \end{pmatrix}, \tag{3}$$

where every subsequent row is a shifted copy of the first row.

Definition 6. The Representer Polynomial

Let $\mathbf{a} = (a_0, a_1, \ldots, a_{n-1})$. Associate with $\mathbf{a}$ the polynomial

$$p_a(z) = a_0 + a_1 z + \cdots + a_{n-1} z^{n-1}. \tag{4}$$

We will call this polynomial the representer of the circulant matrix $C = \operatorname{circ}(\mathbf{a})$.

Since $\mathbf{a}$ uniquely defines a circulant matrix of order n, and the association $\mathbf{a} \leftrightarrow p_a(z)$ is linear, we have that every circulant matrix has a unique representer polynomial.

Definition 7. The Primitive Roots of Unity

Let $n \geq 1$ be a fixed integer and $i = \sqrt{-1}$. The primitive nth roots of unity are

$$\omega_k = e^{\frac{2\pi i k}{n}} = \cos\left(\frac{2\pi k}{n}\right) + i \sin\left(\frac{2\pi k}{n}\right), \qquad k = 0, 1, 2, \ldots, n-1.$$

The right-hand expression follows from applying De Moivre's formula, outlined in many textbooks [1]. We have the following key properties.

1. $\omega^{n-k} = \omega^{-k} \quad \text{for} \quad k = 0, 1, 2, \ldots, n-1.$
2. $\omega^{-1} = \overline{\omega}$
3. $1 + \omega + \omega^2 + \cdots + \omega^{n-1} = 0.$

Definition 8. The Fourier Matrix

Let F_n be the normalized $n \times n$ discrete Fourier transform matrix. That is,

$$F_n = \frac{1}{\sqrt{n}} \begin{pmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & \omega & \omega^2 & \cdots & \omega^{n-1} \\ 1 & \omega^2 & \omega^4 & \cdots & \omega^{2(n-1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \omega^{n-1} & \omega^{2(n-1)} & \cdots & \omega^{(n-1)(n-1)} \end{pmatrix}. \tag{5}$$

Denote the columns of the matrix by $v_0, v_1, \cdots, v_{n-1}$. In general, v_j has the form

$$v_j = \begin{pmatrix} 1 \\ \omega^j \\ \omega^{2j} \\ \vdots \\ \omega^{(n-1)j} \end{pmatrix}, \qquad j = 0, 1, \cdots, n-1.$$

3.2 Supporting Results

3.2.1 Eigenvectors of Circulant Matrices

Let $j \in \{0, 1, \cdots, n-1\}$ and consider a column v_j of the Fourier matrix F. In general, v_j has the form

$$v_j = \begin{pmatrix} 1 \\ \omega^j \\ \omega^{2j} \\ \vdots \\ \omega^{(n-1)j} \end{pmatrix} \tag{6}$$

Let $C = \text{circ}(\mathbf{a})$ be a real circulant matrix like in equation 3. By computation,

$$Cv_j = \begin{pmatrix} a_0 + a_1\omega^j + a_2\omega^{2j} + \cdots + a_{n-1}\omega^{(n-1)j} \\ a_{n-1} + a_0\omega^j + a_1\omega^{2j} + \cdots + a_{n-2}\omega^{(n-1)j} \\ \vdots \\ a_1 + a_2\omega^j + a_3\omega^{2j} + \cdots + a_0\omega^{(n-1)j} \end{pmatrix}$$

Since

$$\omega^{(n-1)j}\omega^{j} = \omega^{nj} = 1,$$

we can retrieve the second entry of Cv_j by multiplying the first entry with ω^j. This is true for any 2 consecutive entries. In general, for $k = 1, 2, \cdots, n-1$, we have that

$$(Cv_j)(k) = \omega^j (Cv_j)(k-1).$$

From the recursive pattern, we recover that

$$(Cv_j)(k) = \omega^{kj} (Cv_j)(0).$$

In vector form,

$$C \begin{pmatrix} 1 \\ \omega^j \\ \omega^{2j} \\ \vdots \\ \omega^{(n-1)j} \end{pmatrix} = (Cv_j)(0) \cdot \begin{pmatrix} 1 \\ \omega^j \\ \omega^{2j} \\ \vdots \\ \omega^{(n-1)j} \end{pmatrix},$$

or alternatively,

$$Cv_j = (Cv_j)(0) \cdot v_j.$$

We note that

$$\begin{aligned} (Cv_j)(0) &= a_0 + a_1\omega^j + a_2\omega^{2j} + \cdots + a_{n-1}\omega^{(n-1)j} \\ &= p_a(\omega^j) \end{aligned}$$

So for $j = 0, 1, \ldots, n-1$, v_j is an eigenvector of C with eigenvalue $p_a(\omega^j) = (Cv_j)(0)$.

Consider 2 distinct eigenvectors v_l and v_m for $l, m \in \{0, 1, \cdots, n-1\}$. Assume without loss of generality that $l > m$. We consider the standard inner product on $\mathbb{C}^n$ and use the properties of the nth roots of unity detailed in Definition 7 of Section 3.1.

$$\begin{aligned} \langle v_l, v_m \rangle &= \sum_{k=0}^{n-1} v_l(k)\overline{v_m(k)} \\ &= \sum_{k=0}^{n-1} \omega^{kl}\omega^{-km} \\ &= \sum_{k=0}^{n-1} (\omega^{l-m})^k = 0 \end{aligned}$$

So any 2 columns of F are orthogonal to each other. We can conclude that the columns of F are orthogonal eigenvectors of C with eigenvalues $\lambda_j = p_a(\omega^j)$ for $j = 0, 1, \cdots n-1$.

3.2.2 Properties of Eigenvalues

Let $C = \text{circ}(\mathbf{a})$ now be a real symmetric circulant matrix. We can show that for $j = 1, 2, \ldots, n-1$,

$$\lambda_j = \lambda_{n-j}.$$

Recall from Section 3.1 that

$$p_a(z) = a_0 + a_1 z + \cdots + a_{n-1} z^{n-1}.$$

We showed above that

$$\lambda_j = p_a(\omega^j) = a_0 + a_1\omega^j + a_2\omega^{2j} + \cdots + a_{n-1}\omega^{(n-1)j}.$$

By property of the nth roots of unity

$$\omega^{n-j} = \omega^{-j} = \overline{\omega^j}.$$

Then

$$\begin{aligned}
\lambda_{n-j} &= p_a(\omega^{n-j}) \\
&= a_0 + a_1\omega^{n-j} + a_2\omega^{2(n-j)} + \cdots + a_{n-1}\omega^{(n-1)(n-j)} \\
&= a_0 + a_1\overline{\omega^j} + a_2\overline{\omega^{2j}} + \cdots + a_{n-1}\overline{\omega^{(n-1)j}}.
\end{aligned}$$

The coefficients of $\mathbf{a}$ are all real, so by linearity of complex conjugation,

$$\begin{aligned}
\lambda_{n-j} &= \overline{a_0 + a_1\omega^j + a_2\omega^{2j} + \cdots + a_{n-1}\omega^{n-1}} \\
&= \overline{\lambda_j}
\end{aligned}$$

Since C is real and symmetric, all of its eigenvalues are real (see for example [7]). It follows that

$$\lambda_{n-j} = \overline{\lambda_j} = \lambda_j$$

for all $j = 1, 2 \cdots, n-1$.

3.2.3 Diagonalisation of Circulant Matrices

As a corollary of the results outlined above, we have the following theorem, first proposed by Philip Davis in [3].

Theorem. (Davis)

If $C = \text{circ}(\mathbf{a})$ is a circulant matrix, it is diagonalizable by the Fourier matrix F. More precisely,

$$C = F^* \, \Lambda_C \, F$$

where

$$\Lambda_C = \text{diag}\,(\, p_a(1),\, p_a(\omega),\, \ldots,\, p_a(\omega^{n-1})\,)$$

Here $p_a(z)$ is the representer polynominal of C.

This is a special case of the Spectral Theorem, which states that for every real, symmetric matrix M, there exists a unitary matrix U and a diagonal matrix D such that $M = U^*DU$. In particular, the diagonal entries of D are the eigenvalues of M and the columns of U are the eigenvectors of L.

3.3 The Laplacian Eigenvectors of Circulant Graphs

Let L be the Laplacian matrix of the general circulant graph with degree $2d$. Let

$$\mathbf{c} = (2d, -1, \ldots, -1, 0, \ldots, 0, -1, \ldots, -1)$$

where -1 is repeated a total of $2d$ times. Then

$$L = \text{circ}(\mathbf{c}).$$

We are interested in the eigenvectors corresponding the 2nd and 3rd *smallest* eigenvalues of the matrix L. In order to deduce which columns of F_n correspond to these eigenvalues, we want to find $j \in \{0, 1, \ldots, n-1\}$ such that

$$\lambda_j = p_c(\omega^j) \tag{7}$$

is minimal.

We have that

$$\begin{aligned}
\lambda_j = p_c(\omega^j) &= 2d - \sum_{k=1}^{d} (\omega^{jk} + \omega^{j(n-k)}) = 2d - \sum_{k=1}^{d} (\omega^{jk} + \omega^{-jk}) \\
&= 2d - \sum_{k=1}^{d} (\omega^{jk} + \overline{\omega^{jk}}) = 2d - \sum_{k=1}^{d} 2 \cdot \text{Re}(\omega^{jk}) \\
&= 2d - 2\sum_{k=1}^{d} \cos(\frac{2\pi jk}{n})
\end{aligned}$$

Every eigenvalue λ_j is associated with some angle $\theta_j = \frac{2\pi j}{n}$ such that

$$\lambda_j = 2d - 2\sum_{k=1}^{d} \cos(k\theta_j)$$

In order to minimize λ_j, we want to maximize the sum

$$\sum_{k=1}^{d} \cos(k\theta_j).$$

The first eigenvalue of a Laplacian matrix is always 0. This corresponds to the case $j = 0$. We are interested in the eigenvectors corresponding to the two smallest nonzero eigenvalues.

The Laplacian matrix L is a real symmetric circulant matrix, so using the result from Section 3.2.2, it's eigenvalues satisfy

$$\lambda_j = \lambda_{n-j} \qquad \text{for } j = 1, \cdots, n-1.$$

Consequently, to analyze the nonzero eigenvalues, it is sufficient to consider indices in the range $1 \leq j \leq \lfloor \frac{n}{2} \rfloor$.

Fix $k = 1$ for now. For $1 \leq j \leq \lfloor \frac{n}{2} \rfloor$, we have that $\theta_j \in (0, \pi]$. The cosine function is decreasing

in the range $(0, \pi]$, so the eigenvalues are naturally ordered from smallest to largest by their indices $1 \leq j \leq \lfloor \frac{n}{2} \rfloor$, and thus correlate directly with the order of the columns of the Fourier matrix. By symmetry, these eigenvalues are reflected in the range $j = \lfloor \frac{n}{2} \rfloor + 1, \ldots, n-1$.

Now let $k \in \{2, 3, \ldots, d\}$. Multiplying every angle $\theta_1, \theta_2, \ldots, \theta_{n-1}$ by an integer k changes the frequency of the $\cos(\theta_j)$ function but does not affect the values of j for which the function takes a maximum value. This preserves the structure of the eigenvalues we outlined above. To conclude,

$$\lambda_2 = \lambda_3 = p_c(\omega^1) = 2d - \sum_{k=1}^{d} \cos(\frac{2\pi k}{n})$$

The corresponding eigenvectors ψ_2, ψ_3 are the columns of the Fourier matrix corresponding to $j = 1$ and $j = n-1$. These are the second column and the last column.

$$\psi_2 = \frac{1}{\sqrt{n}} \begin{pmatrix} 1 \\ \omega \\ \omega^2 \\ \vdots \\ \omega^{n-1} \end{pmatrix} \tag{8}$$

$$\psi_3 = \frac{1}{\sqrt{n}} \begin{pmatrix} 1 \\ \omega^{n-1} \\ \omega^{2(n-1)} \\ \vdots \\ \omega^{(n-1)(n-1)} \end{pmatrix} = \frac{1}{\sqrt{n}} \begin{pmatrix} 1 \\ \omega^{-1} \\ \omega^{-2} \\ \vdots \\ \omega^{-(n-1)} \end{pmatrix} \tag{9}$$

Since L is a real symmetric matrix, we can find a set of n orthogonal eigenvectors that are all real. The eigenvalue λ_1 has multiplicity 2 since $\lambda_1 = \lambda_2$ and $\lambda_1 \neq \lambda_j$ for all $j = 3, 4, \ldots n-1$. Therefore, the orthogonal eigenvectors ψ_2 and ψ_3 form a basis for the eigenspace E_{λ_1} associated with λ_1. Thus, any linear combination of the 2 vectors also belongs to said eigenspace. We combine the vectors to find a pair of real eigenvectors in E_{λ_2}:

$$\psi_2 + \psi_3 = \frac{1}{\sqrt{n}}\left(\begin{pmatrix} 1 \\ \omega \\ \omega^2 \\ \vdots \\ \omega^{n-1} \end{pmatrix} + \begin{pmatrix} 1 \\ \omega^{-1} \\ \omega^{-2} \\ \vdots \\ \omega^{-(n-1)} \end{pmatrix} \right) = \frac{1}{\sqrt{n}} \begin{pmatrix} 1+1 \\ \omega + \overline{\omega} \\ \omega^2 + \overline{\omega^2} \\ \vdots \\ \omega^{n-1} + \overline{\omega^{n-1}} \end{pmatrix} = \frac{2}{\sqrt{n}} \mathrm{Re} \begin{pmatrix} 1 \\ \omega \\ \omega^2 \\ \vdots \\ \omega^{(n-1)} \end{pmatrix}.$$

Similarly one can derive

$$\psi_2 - \psi_3 = \frac{2i}{\sqrt{n}} \cdot \mathrm{Im} \begin{pmatrix} 1 \\ \omega \\ \omega^2 \\ \vdots \\ \omega^{(n-1)} \end{pmatrix}.$$

We can now define two real eigenvectors with eigenvalue λ_2.

$$\phi_2 = \frac{\psi_2 + \psi_3}{2} = \frac{1}{\sqrt{n}} \cdot \mathrm{Re} \begin{pmatrix} 1 \\ \omega \\ \omega^2 \\ \vdots \\ \omega^{(n-1)} \end{pmatrix}$$

$$\phi_3 = \frac{\psi_2 - \psi_3}{2i} = \frac{1}{\sqrt{n}} \cdot \mathrm{Im} \begin{pmatrix} 1 \\ \omega \\ \omega^2 \\ \vdots \\ \omega^{(n-1)} \end{pmatrix}$$

Since ϕ_2 corresponds to the real part and ϕ_3 corresponds to the imaginary part of the primitive nth roots of unity, plotting their coordinates in a 2-dimensional plane traces a circle in the complex plane. In particular, if we consider a sequence of points

$$v_k = (\phi_2(k), \phi_3(k)) \quad \text{for } k = 0, 1, \ldots, n-1$$

the point v_k corresponds to the point $\frac{1}{\sqrt{n}}\omega^{\frac{2\pi}{n}k}$, represented in the complex plane.

To explicitly express the real eigenvectors of λ_2, we recall the identity $\omega^k = \cos\frac{2\pi k}{n} + i\sin\frac{2\pi k}{n}$. So

$$\mathrm{Re}(\omega^k) = \cos\frac{2\pi k}{n}, \quad \mathrm{Im}(\omega^k) = \sin\frac{2\pi k}{n}.$$

The entries of ϕ_2 and ϕ_3 thus correspond to the cosines and sines of the same sequence of angles.

$$\phi_2 = \begin{pmatrix} \cos 0 \\ \cos\frac{2\pi}{n} \\ \cos\frac{4\pi}{n} \\ \vdots \\ \cos\frac{2\pi(n-1)}{n} \end{pmatrix}, \quad \phi_3 = \begin{pmatrix} \sin 0 \\ \sin\frac{2\pi}{n} \\ \sin\frac{4\pi}{n} \\ \vdots \\ \sin\frac{2\pi(n-1)}{n} \end{pmatrix} \tag{10}$$

Therefore, when plotted in the complex plane, the points v_k form a set of evenly spaced points along the unit circle. □

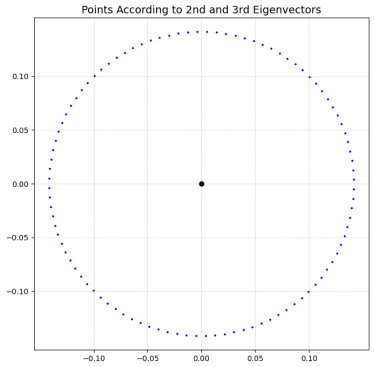


Figure 3: Embedding of a circulant graph, $n = 100$

The circular embedding structure is a property of every circulant graph. By Section 2.1, this layout is an optimal placement of the vertices in the 2-dimensional plane. We previously introduced the matrix of a general circulant graph, in which all the non-zero entries are within a diagonal region. When we use graphs to model real-world networks, they rarely fit this distinct shape, and will often include some degree of randomness. It is, therefore, interesting to study how well this property persists when we introduce randomness into the circulant structure.

In the following sections, we discuss three different ways in which we can generate random circulant graphs and examine how well each model preserves the circular embedding property.

4 Random Circulant Graphs

We can generate a random circulant graph $G = (V, E)$ by removing some of its edges with uniform probability.

Uniform Placement Model

1: Place n points uniformly around the unit circle. This will be our set of vertices V.

2: Form a circulant graph by connecting each vertex to some constant d number of neighbours on each side. This will be our set of edges E.

3: Fix $p \in [0, 1]$. For every edge $e \in E$, remove the edge from the graph with probability $1 - p$.

4: Return $G = (V, E)$

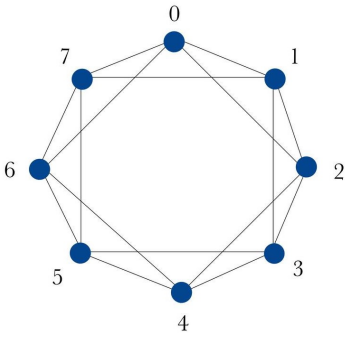
4.1 The Model Matrix

Let H be the adjacency matrix of the original circulant graph with no edges removed.

This model can be represented by a model matrix M where an entry M_{ij} corresponds to the probability that there is an edge between the vertices v_i and v_j.

$$M_{ij} = P(v_i \sim v_j) = \begin{cases} p & \text{if } H_{ij} = 1, \\ 0 & \text{otherwise.} \end{cases}$$

By ordering the vertices according their order along the circle, the model matrix is of the form



$$M = pH = p \begin{bmatrix} 0 & 1 & 1 & 0 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 & 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 & 1 & 1 & 0 & 1 \\ 1 & 1 & 0 & 0 & 0 & 1 & 1 & 0 \end{bmatrix}$$

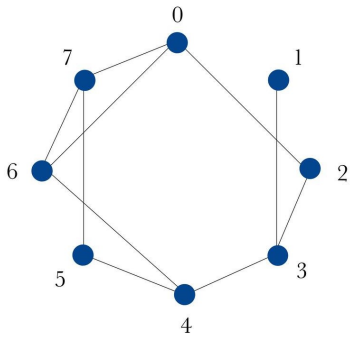
We can generate a random circulant graph G with adjacency matrix S according to our model by sampling edges with the probabilities given in M. The entries in S correspond to independent Bernoulli variables, where $P(S_{ij} = 1) = M_{ij}$.

$$S_{ij} = \begin{cases} 1 & \text{with probability } M_{ij}, \\ 0 & \text{otherwise.} \end{cases}$$

The expectation of the size of a neighbourhood of any vertex u in such a graph is

$$E(|N_G(u)|) = p \times |N_H(u)| = p \times 2d \tag{11}$$

The resulting random graph and associated adjacency matrix could, for example, look like



$$S = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 1 & 1 & 0 \end{bmatrix}.$$

We note that S remains symmetric since $S_{ij} = S_{ij}$. The matrix is, however, no longer circulant. Since entries were randomly removed from the circulant model matrix, the rows are no longer identical shifted copies of the first row. Given such a matrix S, it is not always evident that it was sampled from a circulant model graph. In the example matrix, by construction, the vertices of the random graph G are ordered the same way as in the model graph H. When dealing with real-world network data, however, these labels are not available in advance.

When we discuss circulant graphs, we will often refer to the natural circulant order of vertices. The natural order is to label the vertices from 0 to $n-1$ as we travel along the graph's cycle such that a vertex i is connected to vertices $|i - s| \mod n$ for every $s = 1, \ldots d$. As shown in Section 3, placing the vertices of a circulant graph along the perimeter of a circle in this natural order results in an

embedding minimizing the sum of their squared distances. If we then remove edges from the graph randomly, we are perturbing the structure of the graph, but the underlying truth about the original order of the vertices still remains.

Imagine we are given a random graph G with adjacency matrix S generated by the described model. We are interested in retrieving the model matrix M from which it was sampled from. Uncovering the underlying model can give insight into the structures that generated the graph.

4.2 Recovering the Circulant Layout of Random Graphs

The authors in [8] have found an algorithm that takes a random graph G with adjacency matrix S generated according to the model introduced above and returns the underlying circulant ordering of vertices with a bounded error. The algorithm uses the eigenvectors corresponding to the second and third largest eigenvalues of S.

Algorithm 1 (Richter and Rocha, 2018)

Require: Random matrix S
1: Compute $\hat{x}$ and $\hat{y}$, the eigenvectors for $\lambda_2(S)$ and $\lambda_3(S)$
2: Compute the angular coordinate θ_i for the point of coordinates $(\hat{x}_i, \hat{y}_i)$
3: Define a permutation σ such that $\sigma(i) > \sigma(j)$ iff $\theta_i \geq \theta_j$
4: **return** σ

The permutation σ returned by the algorithm is the suggested order of the vertices of the graph around the circle. The authors choose to quantify the error by the Kendall distance between the original ordering of the vertices and the ordering returned by their algorithm.

The Kendall distance is defined as the number of pairs of vertices for which their relative order has been inverted by the permutation.

$$D(\sigma) = |\{(i, j) : \sigma(j) < \sigma(i) \text{ and } i < j\}|$$

The total number of pairs of vertices in a graph of order n is $O(n^2)$. In their paper, Richter and Rochta show that with high probability,

$$\frac{D(\sigma)}{n^2} \longrightarrow 0 \qquad \text{as} \qquad n \longrightarrow \infty.$$

In practice, this means that the proportion of inverted pairs of vertices goes to zero as n grows large. The success of the algorithm does not depend on the probability p with which we remove edges randomly from the model graph. In the following section we will do a brief overview of the main arguments supporting the result.

4.3 Results on Perturbation Bounds

The embedding of the model matrix M and the embedding of the sampled matrix S will be similar as n grows large.

To bound the error between M and S, we introduce a perturbation matrix R such that

$$R = S - M$$

By this construction, the entries R_{ij} have the following structure:

$$R_{ij} = S_{ij} - M_{ij} = S_{ij} - P(S_{ij} = 1)$$

$$R_{ij} = \begin{cases} 0 & \text{if } M_{ij} = 0 \\ 1-p & \text{with probability } p \\ p & \text{with probability } 1-p \end{cases}$$

We now introduce a lemma from [11]. Here $||\cdot||$ denotes the spectral norm.

Lemma 1. (Norm of a random matrix)

There exists a constant $C > 0$ such that the following holds. Let E be a symmetric matrix whose upper diagonal entries e_{ij} are independent random variables where $e_{ij} = 1 - p_{ij}$ or $-p_{ij}$ with probabilities p_{ij} and $1 - p_{ij}$, respectively, where $0 \leq p_{ij} \leq 1$. Let $\sigma^2 := \max_{ij} p_{ij}(1 - p_{ij})$. If $\sigma^2 \geq C \log(n)/n$, then

$$P(||E|| \geq C\sigma n^{1/2}) \leq n^{-3}. \tag{12}$$

We notice that by our construction, we can apply Lemma 1 to our perturbation matrix R. It follows that

$$P(||R|| \leq C\sigma n^{1/2}) \geq 1 - n^{-3}$$

for some constant $C > 0$ and $\sigma = \sqrt{p(1-p)}$.

This implies that, with high probability, we can bound the norm of the perturbation between the model matrix and the sampled matrix by something small.

By Weyl's inequality (found in for example [6]), we have the following result on the spectral stability of our perturbed matrix:

$$|\lambda_i(S) - \lambda_i(M)| \leq ||R||.$$

Combining these 2 results, we recover that with high probability, one can bound the deviation of the eigenvalues between S and M by $O(n^{1/2})$. We can also bound the difference between subspaces spanned by the second and third eigenvectors of S and M.

Let $(\lambda_i, \psi_i), (\mu_i, \phi_i)$ be corresponding pairs of eigenvalues and eigenvectors of M and S, respectively. Let θ_i be the smallest angle between vectors $\pm\psi_i$ and $\pm\phi_i$. The Davis-Kahan theorem [2] says that the angular difference θ_i between eigenvectors ψ_i and ϕ_i will be small if the norm of the perturbation matrix R is small in comparison to the gap between λ_i and the other eigenvalues of M.

$$\sin 2\theta_i \leq \frac{2||R||}{\min |\lambda_i - \lambda_j|}.$$

Since $\lambda_2 = \lambda_3$, this version of Davis-Kahan is not well-suited for our specific problem. We can, however, use a variant of the theorem introduced in [12].

Theorem. (Variant of Davis-Kahan Theorem)

Let $M, \hat{M} \in \mathbb{R}^{n\times n}$ be symmetric matrices with eigenvalues $\mu_1 \geq \cdots \geq \mu_n$ and $\hat{\mu}_1 \geq \cdots \geq \hat{\mu}_n$, respectively. Let μ_i and $\hat{\mu}_i$ have corresponding unitary eigenvectors v_i and $\hat{v}_i$. Let $\min(\mu_1 - \mu_2, \mu_3 - \mu_4) > 0$, define $V = [\ v_2\ \ v_3\ ]$ and $\hat{V} = [\ \hat{v}_2\ \ \hat{v}_3\ ]$. Let Θ be a 2×2 diagonal matrix whose diagonal contains the principal angles between the subspaces spanned by the columns of V and $\hat{V}$. Then

$$||\sin\Theta||_F \leq \frac{2\min(\sqrt{2}||M - \hat{M}||, ||M - \hat{M}||_F)}{\min(\mu_1 - \mu_2, \mu_3 - \mu_4)}.$$

Using this theorem, we can bound the angle between the subspaces spanned by the second and third eigenvectors of S and M in terms of the norm of their perturbation matrix R. The circular embedding of the model matrix M is a characteristic of this subspace. As shown above, we can bound $||R||$ with high probability. Therefore, the deviation between the corresponding subspaces of S and M is expected to be small, implying that their properties will be similar. Consequently, we anticipate that the embedding of S will exhibit a similar circulant structure to that of M. We observe this property when generating random graphs according to this model.

4.4 Examples

We generated circulant graphs with parameters $n = 1000, d = 200$, and removed edges with 3 different probabilities. Below are plotted points whose coordinates are entries of their eigenvectors and the Kendall distances between the original ordering and the permutations returned by Algorithm 1.

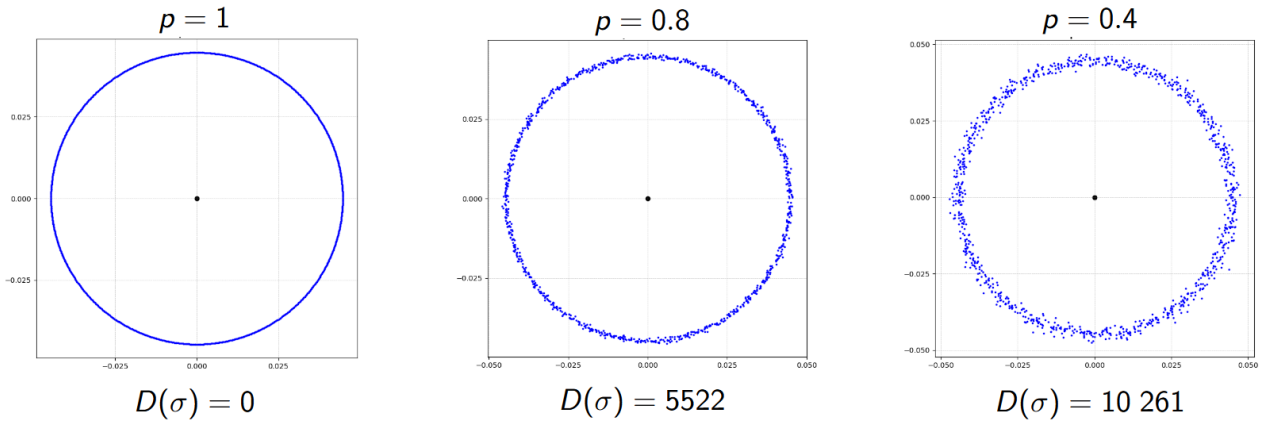


Figure 4: Example embeddings of random graphs with varying edge probabilities, $n = 1000, d = 200$

We notice that when $p = 1$, we recover the ideal circulant embedding. As we decrease the probabilty p that a given edge is in the graph, the points seem to disperse around the ideal circular curve, although the embeddings still exhibit a recognizable circular shape. Decreasing p also increases

the Kendall distance of the embeddings' vertex permutation. The inversion count is comparatively low with respect to the total number of pairs of vertices, of order 10^6.

4.5 Random Circulant Graphs with Uniform Sampling

We have previously considered model graphs in which the vertices are *placed* uniformly around the circle, and each vertex is connected to a fixed number of neighbours. We will now modify this approach slightly, and instead consider a graph in which the vertices are *sampled* uniformly around the circle. That is, we sample n angular coordinates from a uniform distribution over the range $[0, 2\pi)$ and plot points on the unit circle according to these angular coordinates.

We define the angular distance d_θ between two points on a circle to be the absolute difference between their angular coordinates in radians.

Uniform Sampling Model

1: Sample n points uniformly around the unit circle. This will be our set of vertices V.

2: Fix $p \in [0, 1]$. Add an edge between 2 vertices with probability p iff the angular distance between them is below a given threshold distance γ.

3: Return $G = (V, E)$

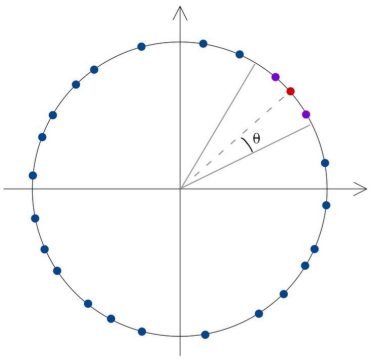


Figure 5: Uniform sampling model

This model can also be represented by a model matrix M where an entry M_{ij} corresponds to the probability that there is an edge between the vertices v_i and v_j.

$$M_{ij} = P(v_i \sim v_j) = \begin{cases} p & \text{if } d_\theta(v_i, v_j) < \gamma, \\ 0 & \text{otherwise.} \end{cases}$$

We sample edges based on the probabilites in M:

$$S_{ij} = \begin{cases} 1 & \text{with probability } M_{ij}, \\ 0 & \text{otherwise.} \end{cases}$$

In contrast to the previously defined uniform vertex placement model, the model matrix M of the uniform sampling model is no longer circulant itself. The 2nd and 3rd eigenvectors no longer have a coherent sinusoidal shape and the results outlined in the previous section are not applicable. We can, however, expect graphs generated by the sampling model to probabilistically behave similarly to the placement model. If we sample n points independently from a uniform distribution on the unit circle, the expectation is that they will be evenly spaced. Since the threshold distance is kept constant along the circle, every vertex would be connected to a constant amount of neighbours. Connecting any 2 such neighbours with probability p is equivalent to removing an edge with probability $1 - p$.

For any vertex u, the fraction of the circle that lies within the threshold distance is $\frac{\gamma}{\pi}$. So

$$E(|N(u)|) = p \times (n-1) \times \frac{\gamma}{\pi},$$

where n is the order of G. This graph behaves similarly to a graph generated by the uniform vertex placement model where every vertex in graph is initially connected to $\frac{n\gamma}{\pi}$ neighbours.

We observe the following embeddings when generating random graphs according to this model.

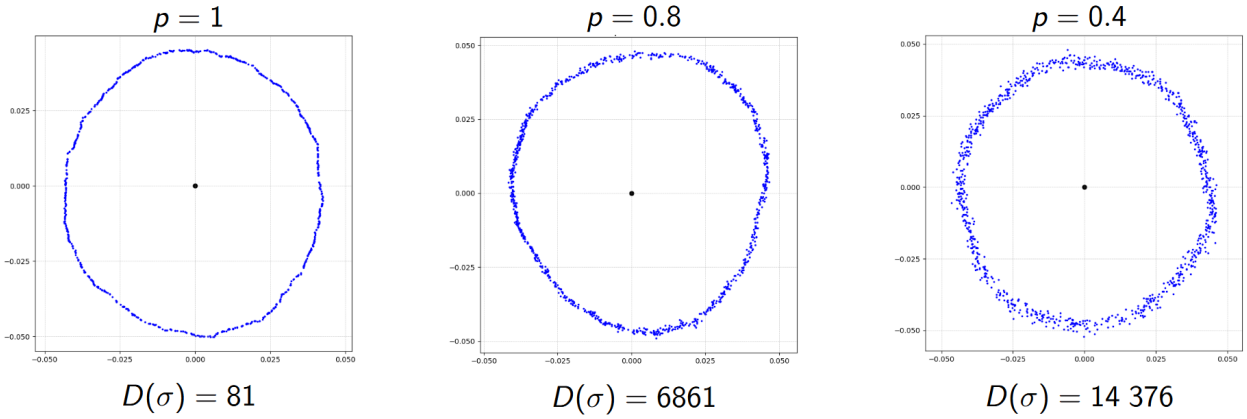


Figure 6: Example embeddings of uniformly sampled graphs, $n = 1000, \gamma = 72°$

The threshold degree $\gamma = 72°$ was chosen to correspond a random circulant graph where before removing edges, every vertex is connected to 200 neighbours. These were the parameters used in the example simulations for the previous random graph model in Fig. 4.

We notice that even when $p = 1$ and we connect a vertex to every vertex within the threshold degree γ, the embedding is not quite circular. Due to the probabilistic nature of the vertex sampling method, some parts of the graph may have vertices more densely packed together, while other parts may have vertices more spread out. As a result, the number of vertices within the threshold distance of a vertex is no longer constant over the entire vertex set. The Laplacian matrix is no longer circulant,

and therefore, the 2nd and 3rd eigenvectors do not exhibit the distinct sinusoidal structure described in eq. 10.

As we observed with the previous random graph model, decreasing the probability p that an edge is added between 2 vertices disturbs the structure of the embedding, causing the points to spread out around the ideal circular curve. In addition, decreasing p increases the Kendall distance of the embedding's vertex permutation. The Kendall distance, however, remains small compared to the total number of pairs, of order 10^6.

5 Non-uniformly Dense Random Circulant Graphs

The uniform sampling model introduces some randomness into our circulant graph. We observed that the embedding of the 2nd and 3rd eigenvector still closely resembled that of an ideal circulant graph. We now want to investigate how this property holds up when we alter the uniformity of the graph. To this end, let us consider a slightly modified approach to generating circulant graphs.

Non-uniform Sampling Model

1: Sample $\frac{2n}{5}$ points uniformly from the first quadrant of the unit circle. Sample a total of $\frac{3n}{5}$ points uniformly from the remaining quadrants of the unit circle. These points will be our set of vertices V.

2: Fix $p \in [0, 1]$. Add an edge between 2 vertices with probability p iff the angular distance between them is below a given threshold distance γ.

3: Return $G = (V, E)$

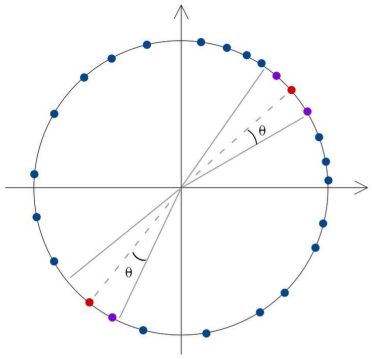


Figure 7: Non-uniform sampling model

5.1 The Adjacency Matrix

This model, similarly to the uniform sampling case, can be represented by a model matrix M where an entry M_{ij} corresponds to the probability that there is an edge between the vertices v_i and v_j.

$$M_{ij} = P(v_i \sim v_j) = \begin{cases} p & \text{if } d_\theta(v_i, v_j) < \gamma, \\ 0 & \text{otherwise.} \end{cases}$$

The expectation of the sampling method is $\frac{2n}{5}$ vertices evenly spaced in the first quadrant, and $\frac{3n}{5}$ vertices evenly spaced along the 3 remaining quadrants. The threshold distance for a possible edge connection is kept constant around the circle. As a result, the vertices in the first quadrant will probabilistically have twice as many vertices in their neighbourhood than those in the rest of the graph. Since the degree of vertices varies, the model matrix M is no longer circulant.

We can generate a random circulant graph with adjacency matrix S by sampling edges according to the probabilities given in M. Since the underlying model matrix is not circulant, we can no longer predict how the eigenvectors of S, and thus their embedding, will behave.

5.2 Randomly Generated Graphs

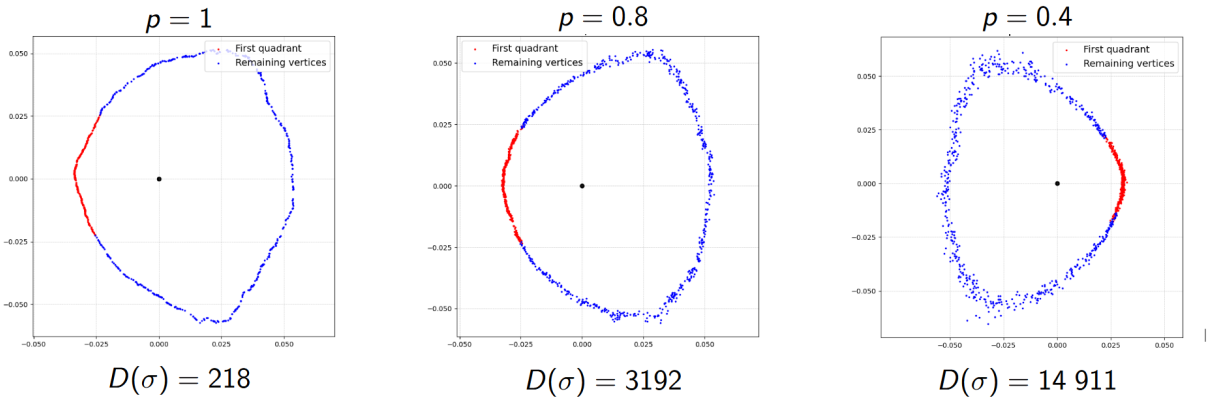


Figure 8: Example embeddings of non-uniformly sampled graphs, $n = 1000, \gamma = 72°$

We generated non-uniformly sampled random graphs with varying edge sampling probabilities. The threshold degree $\gamma = 72°$ was chosen to match the parameters of the previous random graph models in figures 4 and 6. The embedding of the vertices in the first quadrant are highlighted in red. We notice that the embeddings still lie on clear curves around the origin. The curve is no longer centered, and has a denser region closer to the origin. When we decrease the probability that two vertices within the threshold distance are connected, the points seem to disperse and the structure of the embedding is less distinct.

We have empirically observed that the proportion of inverted vertices in the embedding ($\frac{D(\sigma)}{n^2}$) of such random graphs decreases as n increases. We generated 50 instances of each order n graphed below.

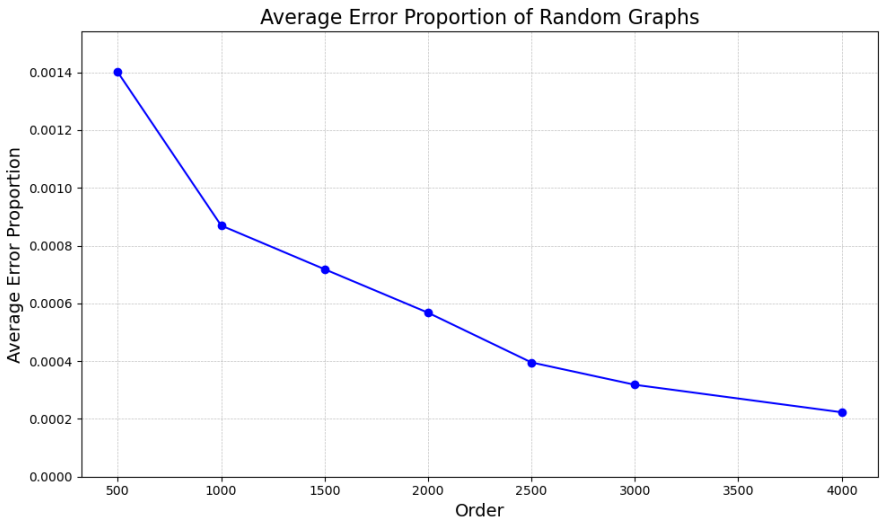


Figure 9: Proportion of inverted pairs of vertices in non-uniformly sampled random graphs

The proportion of inverted pairs of vertices seems to approach 0 as n increases. The algorithm presented in [8] has not yet been proven to recover the underlying circulant structure of such a random graph. Our hypothesis is that the number of inverted pairs of vertices in the ordering returned by the algorithm can be asymptotically bounded by something small.

5.3 Modelling Non-uniform Sampling

To investigate the properties of the generated embeddings, we consider the *expectation* of the non-uniform sampling method. Instead of sampling vertices randomly, we place $\frac{2n}{5}$ vertices uniformly along the first quadrant and $\frac{3n}{5}$ along the rest of the circle. We then connect all vertices within a constant threshold distance γ with edges.

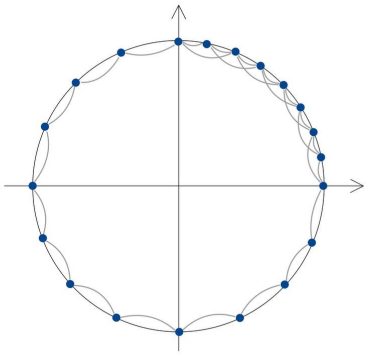


Figure 10: Model non-uniformly dense graph, $n = 20$

We define the vertex set V_1 with the vertices in the first quadrant, and the set V_2 with the rest of the vertices. By construction, the vertices in V_1 will have exactly twice as many neighbours as

vertices in V_2. We can imagine building this graph by replacing a quadrant of a uniformly circulant graph with a twice as dense circulant section. The adjacency matrix of such a graph has the form

$$A = \begin{bmatrix} 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

The denser quadrant (corresponding to the top-left part of the adjacency matrix) will have more edges, and the characteristic diagonal structure of regular circulant graphs is broken. We generated such graphs to analyse their embedding. In the graph below, every vertex in the first quadrant in connected to 4 neighbours. The vertices in the remaining quadrants have the structure of a cycle. This structure is visualised in Fig. 10.

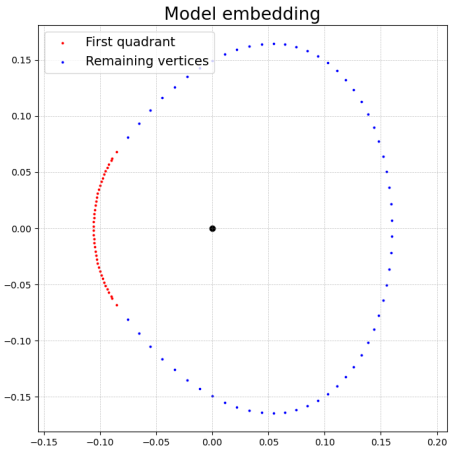


Figure 11: Embedding of the model graph, $n = 100$

We can clearly distinguish a denser region in the embedding closer to the origin. The observed denser region corresponds to the vertices in the denser first quadrant. The rest of the points are spread out more sparsely further from the origin. Since both the 2nd and the 3rd eigenvector are orthogonal to the all ones vector, the sums of each of their entries are necessarily 0. Therefore, we expect that a denser region near the origin has to be compensated for with a less dense region with points further away in the opposite direction.

It is important to note that the points in the embedding are ordered according to the natural circulant ordering of the vertices. This indicates that the algorithm can be successful in retrieving the natural vertex ordering from circulant graphs with a variable density.

We have not yet proven that the algorithm returns a good approximation of the circulant ordering of non-uniformly generated graphs. The proof of the success of the algorithm presented in [8] depends on finding lower bounds for the differences $\lambda_1 - \lambda_2$ and $\lambda_3 - \lambda_4$ of the eigenvalues of the model matrix.

Let C_1 and C_2 denote 2 circulant graphs such that every vertex in C_1 has 2 neighbours and every vertex in C_2 has 4 neighbours. We can retrieve our model graph from Fig. 10 by (a) adding edges to C_1 or (b) removing edges from C_2. By Corollary from [9], we have the following property: Let G be a graph and let H be obtained by adding an edge to G. Then, for all i, $\lambda_i(G) \leq \lambda_i(H)$. In our case this implies that

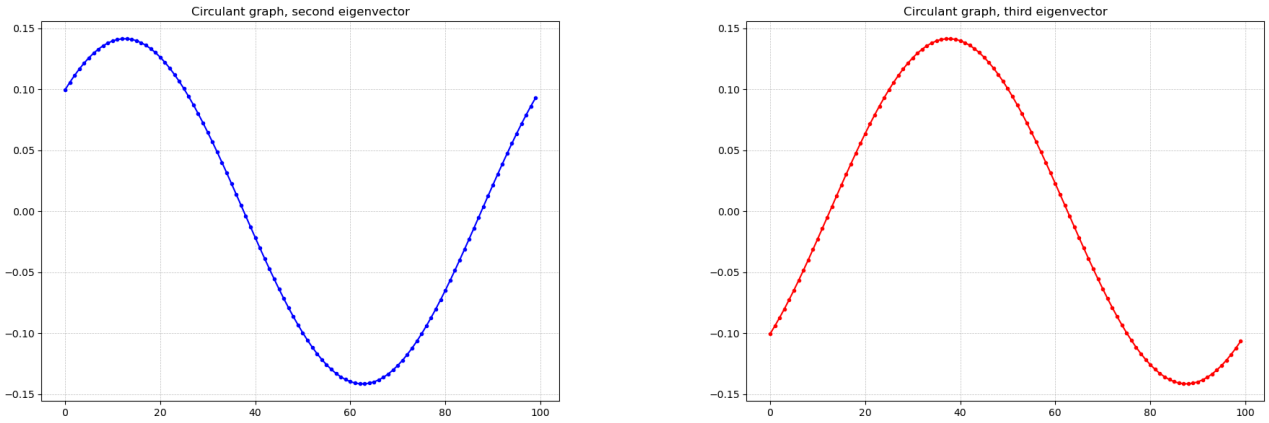
$$\lambda_2(C_1) \leq \lambda_2(M) \leq \lambda_2(C_2)$$
$$\lambda_3(C_1) \leq \lambda_3(M) \leq \lambda_3(C_2).$$

We have empirically observed that the Laplacian eigenvalues of the model graph M fall in between the respective eigenvalues of C_1 and C_2 when all graphs are of the same order n. We have, however, yet to prove a meaningful lower bound for the differences $\lambda_1(M) - \lambda_2(M)$ and $\lambda_3(M) - \lambda_4(M)$.

To gain more insight on the eigenvalues, we focus on examining their corresponding eigenvectors. The embeddings we have presented are, in practice, combining both eigenvectors into one plot. To study their individual structures, we visualised each eigenvector on their own.

6 Eigenvectors of Random Circulant Graphs

Recall that the 2nd and 3rd Laplacian eigenvectors of a circulant graph of order n have an explicit formula: the entries of ϕ_2 and ϕ_3 correspond to the cosines and sines of the same sequence of angles, as described in eq. 10. If we plot the eigenvectors as a function of the vertex index, we recover the uniform sinusoidal structure over one full period.



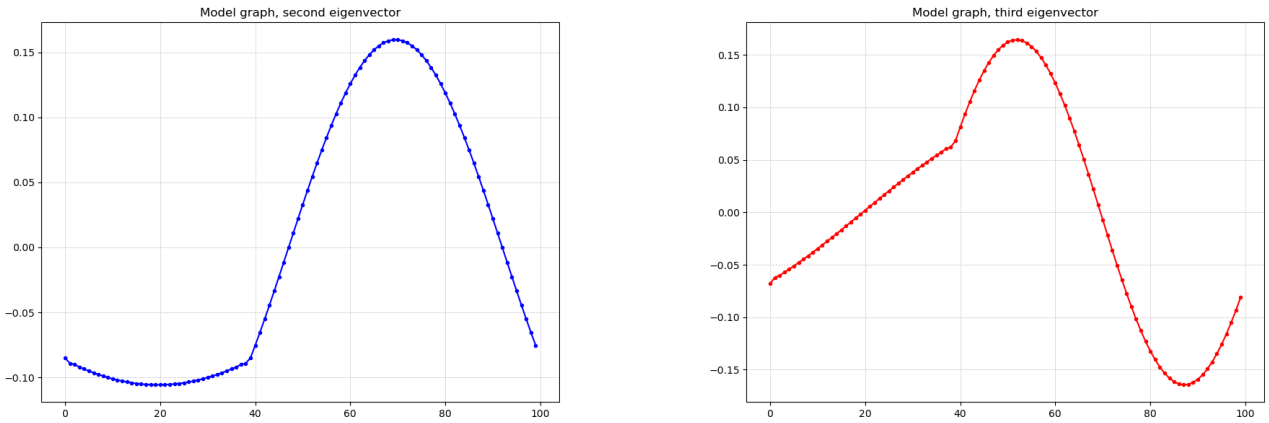
(a) Second eigenvector

(b) Third eigenvector

Figure 12: Laplacian eigenvectors of a circulant graph, $n = 100$

6.1 Non-uniform Sampling Model Graph

To investigate the properties of our theoretical model graph, we plot the 2nd and 3rd eigenvectors individually.

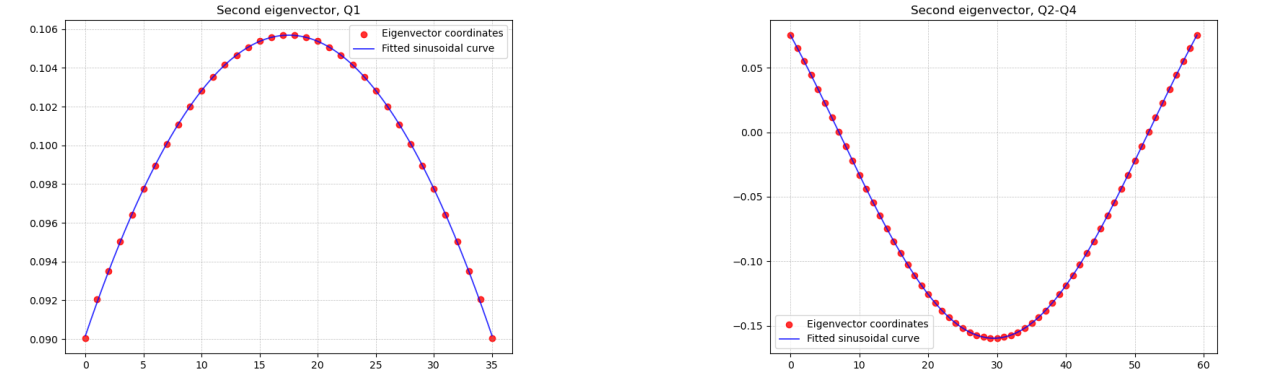


(a) Second eigenvector (b) Third eigenvector

Figure 13: Laplacian eigenvectors of the non-uniformly dense model graph, $n = 100$

We observe that the first and last coordinate are approximately the same, verifying that the vector does one full rotation around the origin. The vector seems to be split into 2 distinct regions. The first region spans vertices 1 - 40; exactly those in the first, denser quadrant of the circle. The second region spans the rest of the vertices 40 - 100. Both regions seem to lie on their own distinct curves. Upon inspection, we hypothesize that we can fit individual sinusoidal curves to each region.

To investigate our hypothesis, we generated a uniform sampling model graph of order 100, computed its 2nd and 3rd eigenvector, and fit individual sinusoidal curves over the dense and sparse regions of the vectors by a non-linear least squares optimization algorithm. We assume that the second eigenvector ϕ_2 is a combination of 2 distinct cosine functions, and that the third eigenvector ϕ_3 is a combination of 2 distinct sine functions. We note that the parameters we are interested in (amplitude and frequency) are identical regardless of whether we fit a cosine or a sine wave over our data. The parameter estimations below were done using SciPy's *curve_fit* function [10], which uses a non-linear least squares optimization to fit a function to the eigenvector coordinates.



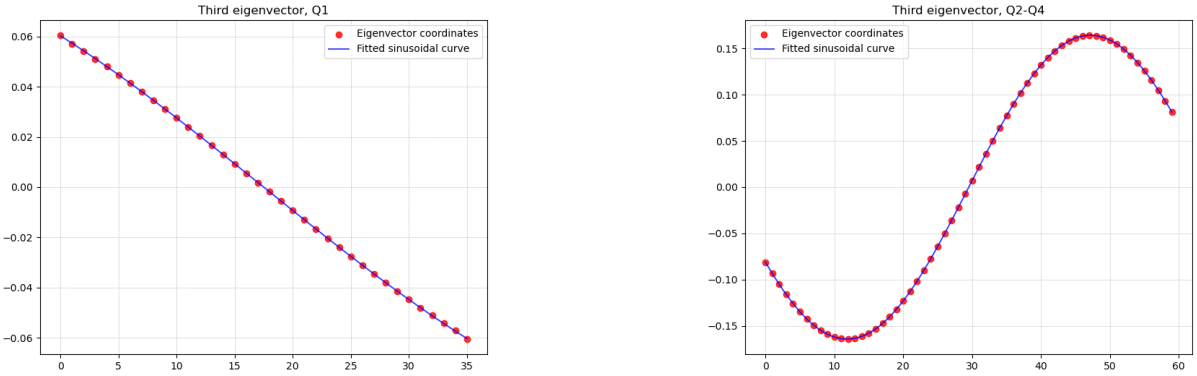
(a) Vertices in the first quadrant (b) Vertices in the remaining quadrants

Figure 14: Sinusoidal fits over 2 regions of the 2nd Laplacian eigenvector of the model graph, $n = 100$

Individually, the different regions of the eigenvectors are very well fitted by cosine functions. The Normalized Mean Squared Error (NMSE) of both curves is 0 up to 6 decimal points, implying the curves are near exact fits. The different regions of the vectors have the following parameters.

$$\phi_2 = \begin{cases} 0.533\,\cos(2\pi\,0.002\,k + \phi_1) + h_1 & \text{for } k \text{ in } V1 \\ 0.160\,\cos(2\pi\,0.011\,k + \phi_2) + h_2 & \text{for } k \text{ in } V2 \end{cases}$$

We repeat these fits for the 3rd eigenvector.



(a) Vertices in the first quadrant

(b) Vertices in remaining quadrants

Figure 15: Sinusoidal fits over 2 regions of the 3nd Laplacian eigenvector of the model graph, $n = 100$

Similarly to the 2nd eigenvector, the curves are very well fit by sine functions. The NMSE is still 0 up to 6 decimal points.

$$\phi_3 = \begin{cases} 0.096\,\sin(2\pi\,0.006\,k + \phi_3) + h_3 & \text{for } k \text{ in } V1 \\ 0.164\,\sin(2\pi\,0.014\,k + \phi_4) + h_4 & \text{for } k \text{ in } V2 \end{cases}$$

To investigate the relationship between the order of the graph and the parameters of the vectors, we generated model graphs of varying orders between 200 and 4000 [2]. We assume that the general structure of the eigenvectors is

$$\phi_2 = \begin{cases} A_1\,\cos(2\pi f_1 k + \phi_1) & \text{for } k \text{ in } V1 \\ A_2\,\cos(2\pi f_2 k + \phi_2) & \text{for } k \text{ in } V2 \end{cases}$$

$$\phi_3 = \begin{cases} A_3\,\sin(2\pi f_3 k + \phi_3) & \text{for } k \text{ in } V1 \\ A_4\,\sin(2\pi f_4 k + \phi_4) & \text{for } k \text{ in } V2 \end{cases}$$

The parameters we wish to obtain are are A_j and f_j for every $j = 1, 2, 3, 4$. We hypothesize that for every such parameter P, there are constants c and α such that $P = cn^\alpha$. To recover the constants, we take the logarithm

$$\log(P) = \log(cn^\alpha) = \log(c) + \alpha\log(n)$$

[2] Beyond 4000 vertices the results become unpredictable, likely due to machine error in computer arithmetic.

and perform an ordinary least squares regression on the transformed data. The parameters were estimated by NumPy's *polyfit* function [5], which returns a least squares polynomial fit.

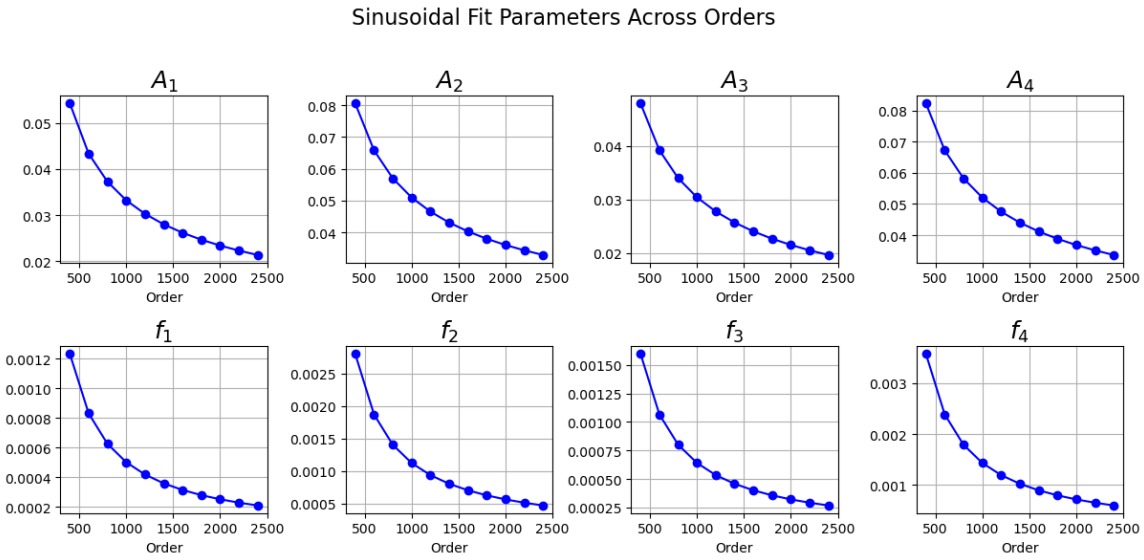


Figure 16: Amplitudes and frequencies fitted curves with respect to order of graph

For visualisation, we plot $\log(P)$ with respect to the independent variable $\log(n)$ individually for every parameter.

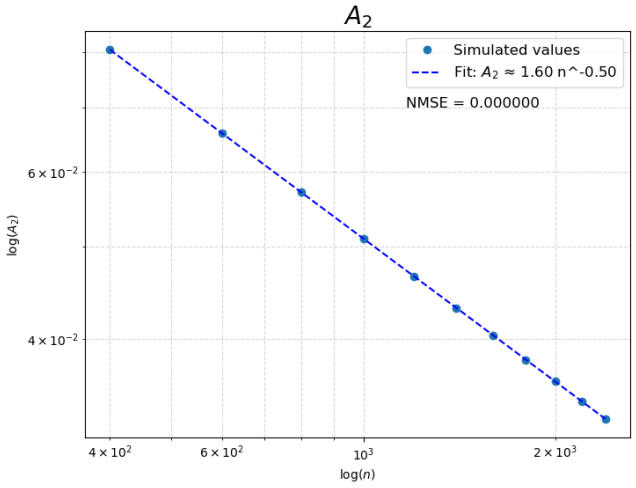


Figure 17: Log of amplitude A_2 with respect to log of order n

The NMSE of the fitting is 0 up to 6 decimal points, implying the fit is nearly exact. From the plot, we recover that the estimated slope α is -0.50 and the estimated y-intercept c is 1.60. We recover the following relationship:

$$A_2 \approx 1.60\, n^{-1/2}.$$

We repeat this for every parameter. The resulting plots are found in Appendix A. The results verify

that the parameters follow a power-law relationship such that

$$A_j \propto n^{-1/2}, \quad f_j \propto n^{-1} \quad \text{for } j = 1, 2, 3, 4.$$

Furthermore, we find the following coefficients:

A_1	A_2	A_3	A_4	f_1	f_2	f_3	f_4
1.13	1.60	0.94	1.65	0.47	1.11	0.63	1.42

As the order of the graph increases, all parameters A_j and f_j decrease according to set power-law relationships, implying these parameters are tightly connected to the graph size. Since the observed relationships between the parameters are identical for every order graph we generated, we can infer that the coefficients are a property of the general structure of the model graph. The coefficients are likely tied to characteristic structures within the model graph, such as the difference in density between the different quadrants and the regular circularity of the edge connections. However, the specific reasoning behind the values of the coefficients is still unclear. Further analysis of the theoretical framework is likely needed to understand their derivation.

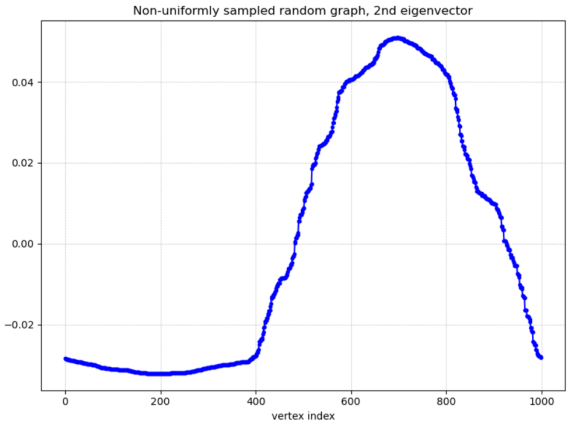
To explore these relationships, we suggest repeating the experiment for modified versions of the model graph. For example,

- Increase the density of the first quadrant by a factor of $3, 4, ...$ with respect to the density of the rest of the graph.
- Increase or decrease the proportion of the dense region in the graph.
- For now our model graph is a cycle with a dense region where every vertex is connected to 4 neighbours. One could experiment with a model graph where every vertex is connected to 4 vertices in the sparse regions, and 8 vertices in the dense region.

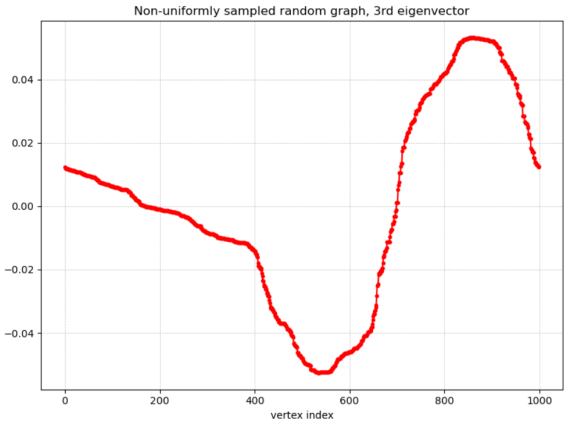
These may offer insight into the structures at play that affect the observed parameters.

6.2 Non-uniformly Sampled Graphs

The goal of creating a theoretical model was to understand where the structures we observe in our non-uniformly sampled random graphs arise from. We repeat the investigation of the 2nd and 3rd Laplacian eigenvector structure for a non-uniformly sampled random graph of order $n = 1000$.



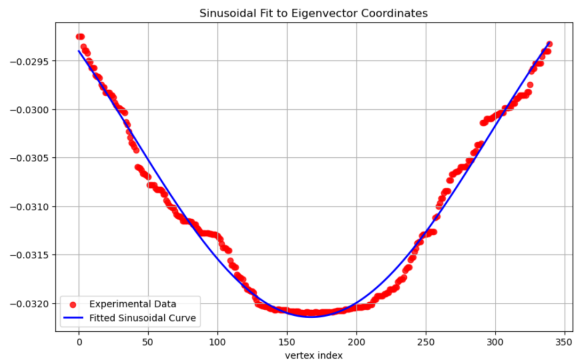
(a) Second eigenvector



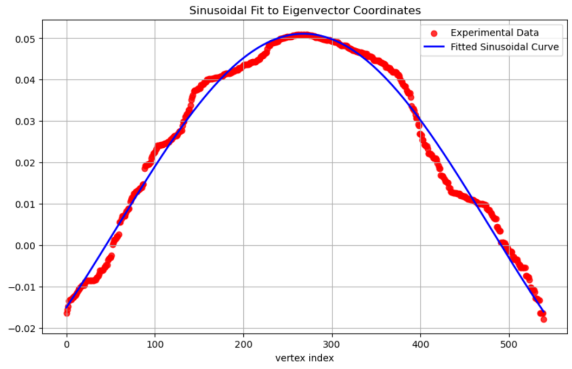
(b) Third eigenvector

Figure 18: Eigenvectors of a non-uniformly sampled random graph, $n = 1000$

We hypothesize that the eigenvectors of graphs generated by the non-uniform random sampling model can be approximated by a combination of two distinct sinusoidal functions. Before fitting data to the two defined regions of the eigenvectors, we remove a fixed number of vertices from both ends of the vectors. Vertices near the borders between the first quadrant and the rest of the circle are more likely to have a higher number of neighbors in the dense quadrant and fewer in the sparser quadrants (either the second or fourth). Such inconsistencies in vertex degree disrupt the clear separation between the two regions. To mitigate this effect, we exclude these border vertices when fitting sinusoidal curves to the eigenvectors.



(a) Dense region



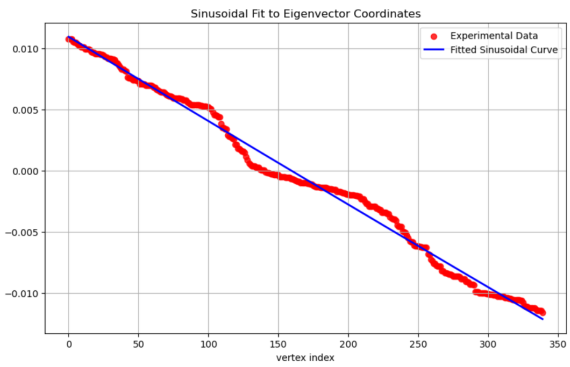
(b) Sparse region

Figure 19: Sinusoidal fit over the 2nd eigenvector of a random graph, $n = 1000$

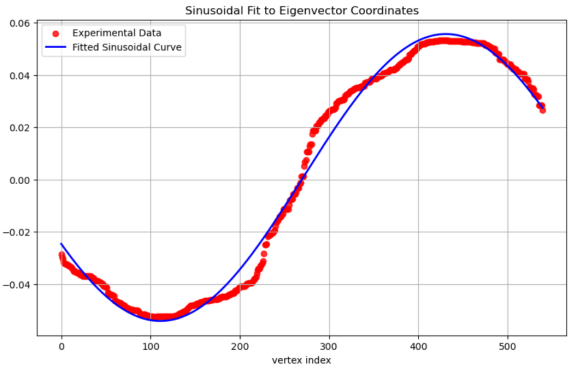
We find the following fitted parameters for the 2nd eigenvector:

$$\phi_2 = \begin{cases} 1.9 \times 10^{-3} \cos(2\pi\, 1.9 \times 10^{-3}\, k + \phi_1) + h_1 & \text{for } k \text{ in } V1 \\ 4.7 \times 10^{-2} \cos(2\pi\, 1.2 \times 10^{-3}\, k + \phi_2) + h_2 & \text{for } k \text{ in } V2 \end{cases}$$

The NMSE values of approximately 0.014 and 0.015 indicate that the fits are decently accurate, suggesting that the sinusoidal model provides a good approximation of the eigenvectors.



(a) Dense region



(b) Sparse region

Figure 20: Sinusoidal fit over the 3rd eigenvector of a random graph, $n = 1000$

We find the following fitted parameters for the 3rd eigenvector:

$$\phi_3 = \begin{cases} 6.2 \times 10^{-1} \sin(2\pi\, 2.2 \cdot 10^{-5}\, k + \phi_3) + h_3 & \text{for } k \text{ in } V1 \\ 5.5 \times 10^{-2} \sin(2\pi\, 1.6 \times 10^{-3}\, k + \phi_4) + h_4 & \text{for } k \text{ in } V2 \end{cases}$$

The NMSE of the fits are approximately 0.010 for both regions, again implying that the sinusoidal model is a suitable option for describing the vectors.

To examine the parameters of the eigenvectors above, we generated a model non-uniform graph of the same order $n = 1000$ and documented the parameters:

Parameter	Model graph	Random graph
A_1	0.03321	0.00191
f_1	0.00050	0.00193
A_2	0.05101	0.04689
f_2	0.00112	0.00118
A_3	0.03042	0.61910
f_3	0.00064	0.00002
A_4	0.05208	0.05490
f_4	0.00143	0.00156

Table 1: Parameters of 2nd and 3rd eigenvectors

We notice that the parameters associated with the sparse regions of the graphs, namely A_2, f_2 and A_4, f_4, differ by at most 10% between the 2 graphs, whereas those corresponding to the dense quadrant can vary by up to an order of magnitude. This suggests that the structure of the eigenvector in sparse regions of non-uniformly sampled random graphs more closely resembles the model graph, while the dense regions exhibit greater variability. The fit for the first region is performed on a smaller number of vertices, yet the NMSE of the sinusoidal fits are of similar magnitude over the 2 different regions.

The parameters in Table 1 correspond to observations from a single instance of a random graph. To strengthen our findings, it would be necessary to repeat the calculations across a large number of generated random graphs.

7 Discussion

Ultimately, we aim to prove that we can recover underlying circulant order from non-uniformly dense random graphs.

7.1 Findings

We observed that Algorithm 1 was still successful in recovering the natural ordering of the vertices of non-uniformly random graphs generated by our sampling model. The proportion of the number of inverted pairs of vertices in the ordering returned by the algorithm decreased as the order of the random graph increased. This suggests that the embedding of our random graph model becomes increasingly organized as the order increases, and that the algorithm seems to be well-suited for recovering the natural ordering. The embedding exhibits a less defined circular structure compared to uniform random graphs, but the vertices were observed to follow a curved orbital pattern around the origin.

The natural ordering of the theoretical non-uniform model graph was exactly recoverable by the algorithm, further supporting its applicability to non-uniform circulant graphs. The 2nd and 3rd Laplacian eigenvectors of the model graph exhibited a distinct structure: the eigenvectors are split into 2 distinct regions that are nearly exactly fitted by sinusoidal functions with known parameters.

It is difficult to verify or predict the parameters of the 2nd and 3rd eigenvectors of randomly generated graphs due to the large variation in the resulting fittings. Our eigenvector fitting method worked well for graphs of moderate size (< 4000). The computational complexity of calculating and interpreting Laplacian eigenvectors increases significantly with the graph size, and makes verifying observed patterns for graphs of larger orders difficult.

7.2 Implications

If we can use the Laplacian eigenvectors to recover the underlying circulant order of a graph, this implies that we can reconstruct the structural properties of random graphs. The ability to recover this structure could lead to a deeper understanding of how nodes in a graph are connected, even when the graph appears random. Knowing the circulant ordering would allow us to predict and reconstruct edge patterns between nodes based on the graph's Laplacian spectrum.

In real-world applications of graphs, such as social, biological, and communication networks, the complete graph structure may be hard to obtain due to the nature of real observations and data collection. Being able to recover underlying structures could help complete missing information and connections when only partial data is available, in essence completing the graph.

In addition, many real-world networks can depend on underlying, latent factors. Recovering the circulant order could help uncover information about hidden factors determining which nodes are connected. For example, in social networks, individuals can often be connected based on underlying, unobservable factors such as shared interests, mutuals, or geographical location, rather than solely through direct connections.

Finding underlying structures can also help us predict which nodes are likely to be connected in

the future. Uncovering periodic structure can help us generalize edge connectivity patterns across different regions of the graph, and thus predict links that are likely to arise from the existing set of connections. For example, consider a scenario in which we want to predict future connections on a social media platform. Instead of predicting solely on the basis of easily inferrable characteristics, such as common connections between two users, using the Laplacian spectrum could take advantage of underlying factors to increase the accuracy of the predicted connections.

From a theoretical perspective, proving this result would provide deeper insights into the relationship between the Laplacian spectrum of graphs and the underlying graph structures. This would facilitate the further study of complex networks in various applied fields.

7.3 Future Work

The natural next step is to formally derive the coefficients of the 2nd and 3rd eigenvectors of the theoretical non-uniform model graph. Knowing the exact structure of the eigenvectors is a crucial step in formally proving that the algorithm is also successful in recovering the underlying order of non-uniformly dense graphs. Continuing experimentation with modified versions of the model graph can help find insight into the factors defining their structure. With a theoretical basis, we would have the tools to extend the approach and account for randomness in the structure.

Acknowledgements

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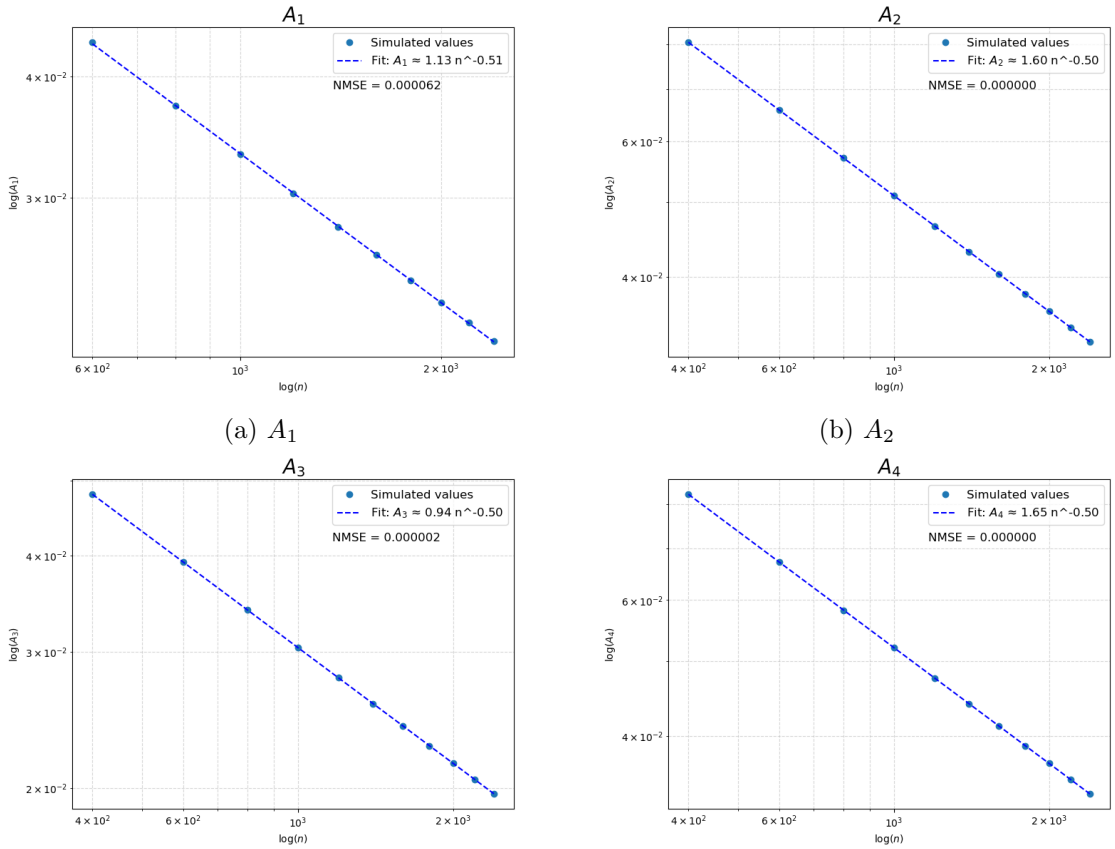
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A Parameter fits

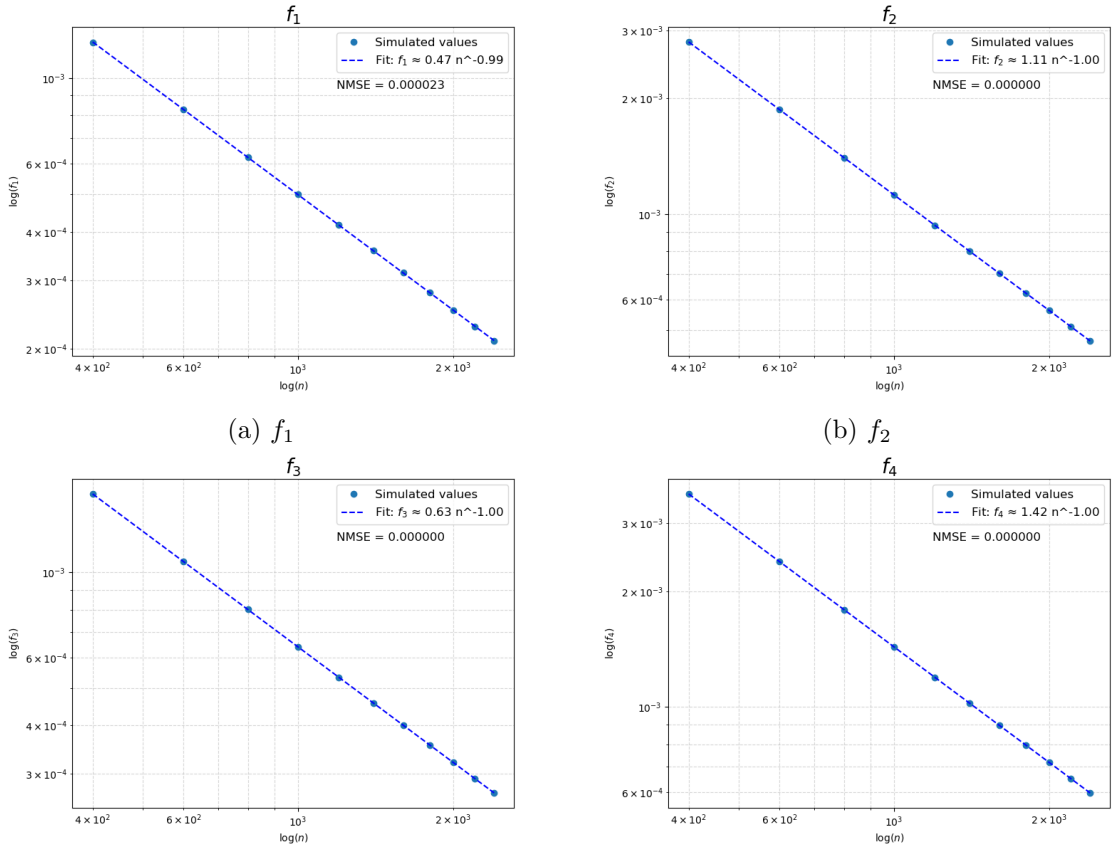
Below we have graphed fits to estimate the parameters of the sinusoidal functions characterizing the 2nd and 3rd eigenvectors of the non-uniform model graph with NumPy's *polyfit* function [5].



(a) A_1 (b) A_2

(c) A_3 (d) A_4

Figure 21: Amplitudes of non-uniform model graph eigenvector functions



(a) f_1 (b) f_2

(c) f_3 (d) f_4

Figure 22: Frequencies of non-uniform model graph eigenvector functions